

Hased

Smart Farm Management System

Project Requirements & Business Rules

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1. Project Overview

Hased is an institutional desktop application designed to manage farm operations. The system provides tools for managing fields, crops, workers, harvests, transactions, and fertilizer/medicine inventory. It serves a single organization (farm) with role-based access for administrators and workers.

Technologies: Java 21, JavaFX, FXML, CSS, PostgreSQL, Maven, BCrypt.

2. System Entities

The database consists of 9 entities:

#	Entity	Description
1	Users	System accounts (Admin and Workers)
2	Farms	Farm information (name, location, size)
3	Farm_Workers	Worker assignments to farms with roles and wages
4	Fields	Individual fields within a farm
5	Crops	Crops planted in each field
6	Fertilizers_Medicines	Fertilizers and medicines applied to fields
7	Farm_Logs	Activity logs (irrigation, plowing, planting, notes)
8	Harvests	Harvest records with good and damaged quantities
9	Transactions	Financial records (sales, purchases, payments)

3. Entity Relationships

All relationships in the system follow a one-to-many (1:N) cardinality pattern:

Entity A	Entity B	Cardinality	Description
Users	Farm_Workers	1 : N	A user can be assigned to multiple farms
Farms	Farm_Workers	1 : N	A farm has multiple workers

Entity A	Entity B	Cardinality	Description
Farms	Fields	1 : N	A farm contains multiple fields
Fields	Crops	1 : N	A field can have multiple crops
Fields	Farm_Logs	1 : N	A field has multiple activity logs
Users	Farm_Logs	1 : N	A user records multiple logs
Fields	Harvests	1 : N	A field produces multiple harvests
Crops	Harvests	1 : N	A crop yields multiple harvests
Users	Harvests	1 : N	A user performs multiple harvests
Fields	Fertilizers_Medicines	1 : N	A field uses multiple fertilizers/medicines
Harvests	Transactions	1 : N	A harvest can generate transactions
Users	Transactions	1 : N	A user can receive payment transactions

4. Business Rules

4.1 User & Authentication Rules

BR-01: The system has exactly one Admin account, created automatically on first launch. It cannot be registered through the Sign Up interface.

BR-02: Each user must provide a name and a password. Either an email or a phone number is required; the other is optional.

BR-03: Email and phone number must be unique across all users.

BR-04: Passwords must be hashed using BCrypt before storage. Plain-text passwords are never stored.

BR-05: Users can log in using either their email or phone number combined with their password.

BR-06: A user's role is either ADMIN or WORKER. Role cannot be changed by the user.

BR-07: Only the Admin can create, modify, or deactivate worker accounts.

4.2 Farm & Field Rules

BR-08: The system manages a single farm (institutional model). The farm record is created on system initialization.

BR-09: A farm can have multiple fields. Each field belongs to exactly one farm.

BR-10: Each field has a name, size (in dunums), location, and soil status.

BR-11: Soil status is categorized as: EXCELLENT, GOOD, FAIR, POOR, or NOT_TESTED.

BR-12: A field cannot be deleted if it has active crops, pending harvests, or scheduled tasks.

4.3 Crop Rules

BR-13: Each crop is planted in exactly one field. A field can contain multiple crops.

BR-14: Crops are categorized by type: VEGETABLE, FRUIT, GRAIN, or TREE.

BR-15: Each crop has a status lifecycle: GROWING → READY → HARVESTED.

BR-16: A crop cannot be deleted if it has associated harvest records.

4.4 Worker & Assignment Rules

BR-17: Workers are assigned to the farm through the Farm_Workers table with a specific job type and wage.

BR-18: Job types are: IRRIGATOR (paid per liter), HARVESTER (paid per kg or piece), and PLOWER (paid per dunum).

BR-19: Each worker assignment has a wage_per_unit and a wage_unit that determines how compensation is calculated.

BR-20: A worker's status is either ACTIVE or INACTIVE. Only active workers can log work or record harvests.

BR-21: A user can have multiple worker assignments (different roles in different contexts).

4.5 Farm Log Rules

BR-22: Farm logs record daily activities: IRRIGATION, PLOWING, PLANTING, FERTILIZING, or general NOTES.

BR-23: Each log entry is linked to a specific field and the user who performed the activity.

BR-24: Logs include a date, optional quantity, and description.

BR-25: Farm logs serve as the basis for automatic task completion detection.

4.6 Harvest Rules

BR-26: Each harvest record is linked to a field, a crop, and the worker who performed it.

BR-27: Harvest records must include both quantity_good and quantity_damaged (damaged can be zero).

BR-28: The unit of harvest is either kg (kilogram) or piece, matching the crop type.

BR-29: Worker compensation for harvesters is automatically calculated as: quantity_good × wage_per_unit.

BR-30: Multiple workers can contribute to harvesting the same crop; each worker's contribution is recorded separately.

4.7 Transaction Rules

BR-31: Transactions are categorized as: SALE (revenue), PURCHASE (expense), or PAYMENT (worker salary).

BR-32: SALE transactions may optionally reference a harvest record (related_harvest_id).

BR-33: PAYMENT transactions must reference the worker receiving payment (related_user_id).

BR-34: All transactions must include an amount, description, and date.

BR-35: Transaction amounts are stored as positive values; the type (SALE/PURCHASE/PAYMENT) determines the financial direction.

4.8 Fertilizer & Medicine Rules

BR-36: Each fertilizer or medicine record is linked to a specific field.

BR-37: Items are categorized as either FERTILIZER or MEDICINE.

BR-38: Each record includes name, quantity, unit, application date, and optional notes.

BR-39: Stock status is tracked: items below a defined threshold are flagged as Low Stock.

4.9 Task & Calendar Rules

BR-40: Tasks are derived from Farm_Logs and Harvests — they are not stored in a separate table.

BR-41: There are two task completion types: 'Any-one' (completed when any single worker finishes) and 'Cumulative' (completed when the combined effort of all workers reaches the target).

BR-42: Irrigation, planting, and fertilizing tasks follow the Any-one model.

BR-43: Harvest tasks follow the Cumulative model: the target quantity must be reached by the sum of all workers' contributions.

BR-44: Task status is determined automatically: Done (matching record exists), Pending (today or future, no record), Missed (past date, no record).

5. Non-Functional Requirements

BR-45: The application runs as a desktop application using JavaFX with FXML and CSS for UI.

BR-46: The database engine is PostgreSQL, accessed through JDBC.

BR-47: The application supports Dark Mode and Light Mode themes, toggleable at any time.

BR-48: Scene transitions preserve window state (maximized, size, position).

BR-49: The system uses BCrypt for password hashing and prepared statements for SQL injection prevention.

BR-50: The UI is responsive: card layouts adapt to window resizing using GridPane with percentage-based column widths.